

Curriculum Vitae — Artur Avakyan

Personal Info

Name: Artur Avakyan

Current position: Post-Doctoral researcher in Tuebingen University

Address: Tübingen, Baden-Württemberg (open to relocation/hybrid)

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ORCID : [0000-0001-7502-7951](https://orcid.org/0000-0001-7502-7951)

Web: [Portfolio](#), [XRBCats](#), : [ScorpionesXone](#)

Professional Summary

Astrophysicist and Software Developer with 5+ years of experience in building complex data pipelines, machine learning models, and high-performance physical engines. Passionate about transitioning from academia to permanent infrastructure development, translating complex astrophysical data into user-friendly web portals and robust backend architectures. Proven ability to lead technical sub-projects within large international collaborations (eROSITA) and deliver tools that serve the global scientific community.

Technical Skills

<i>Programming</i>	Python, C/C++, SQL/ADQL, JavaScript, HTML/CSS
<i>Data Science</i>	Machine Learning, Statistical Modeling, Nested Sampling
<i>Scientific Computing</i>	NumPy, SciPy, pandas, astropy, XSPEC, pyphot, astroARIADNE
<i>Software Engineering</i>	FastAPI, REST APIs, Git, CI/CD, Docker, Testing
<i>Data Engineering</i>	Data Pipelines + Processing + Validation, FITS, Automated Workflows
<i>Scientific Domain</i>	Statistical Inference, MWL Data Analysis, SED Modeling

Work Experience

Scientific Developer / Post- & Doctoral researcher

2021 — Present

Institute for Astronomy and Astrophysics, University of Tübingen, Germany

- **Data Portals & Architecture:** Architected and deployed [XRBCats](#), a comprehensive, standardized relational database and public web portal for Galactic X-ray binaries, providing data access to the global community.
- **Machine Learning & Big Data Pipelines:** Designed and implemented supervised ML pipelines to process, cross-match, and classify millions of sources.
- **Technical Leadership:** Mentored and supervised students in Python development, data analysis, and code optimization.
- **Advanced Computational Modeling:** Expert in Spectral Energy Distribution (SED) fitting and photometric analysis. Applied Bayesian inference and nested sampling algorithms to complex models.

Software Engineer / Researcher

2018 — 2021

Sternberg Astronomical Institute, Moscow State University, Russia

- **High-Performance Computing:** Co-developer of [FREDDI](#), an open-source C++ physical engine for modeling accretion disc dynamics.
- **Numerical Algorithms:** Implemented implicit finite-difference schemes and tridiagonal matrix algorithms to efficiently solve complex non-linear diffusion equations.

Key Software Products & Portfolio

- **XRBCats: Science User Web Portal**
A full-stack project providing a unified catalogue of High- and Low-Mass X-ray Binaries. Used by researchers globally.
- **Interactive 3D Galaxy Visualization**
Developed an interactive, browser-based 3D visualization for sources in the Galaxy
- **FREDDI (C++ Physical Engine)**
Co-authored a fast-rise exponential-decay accretion disc model implementation. Highly optimized C++ code for numerical simulations in astrophysics.

Education

Ph.D. in Astrophysics **2021—2025**

Place: Institut für Astronomie und Astrophysik Tübingen, Universität Tübingen, Germany

Focus: Machine Learning on Big Data, X-ray binary populations, eROSITA survey.

Advisors: Dr. Victor Doroshenko and Prof. Dr. Andrea Santangelo

Master in physics and astronomy **2015—2021**

Place: Sternberg Astronomical Institute, Moscow State University, Russian Federation

Focus: Numerical simulations, accretion disc modelling

Advisors: Dr. Galina Lipunova and Dr. Konstantin Malanchev

Personal Skills

<i>Languages</i>	English (C1+, including professional vocabulary) Russian (native) German (B1+)
<i>Astronomy soft</i>	XSPEC, TOPCAT, proficiency databases such as Vizier/SIMBAD/MAST
<i>Software</i>	Microsoft Office (Word, Excel), GitHub, L ^A T _E X

Service

Member of eROSITA_DE consortium (accreting compact objects and follow-up working groups)

Reviewer in Monthly Notices of the Royal Astronomical Society journal

Reviewer in Journal of High Energy Astrophysics (JHEAP)

Selected Publications

Highlighting papers focused on software products, data engineering, and machine learning pipelines. Full list of publications available on [ORCID](#).

- **Avakyan A.**, Zainab A., Doroshenko V., et al. (upcoming, 2026). *"Hunting down new Galactic XRBs in eROSITA using ML"*.
Tech Focus: Automated classification of millions of X-ray sources using ML on MWL Big Data.
- **Avakyan A.**, Neumann M., Zainab A., Doroshenko V., et al. (2023). *"XRBCats: Galactic Low-Mass X-ray binary catalogue"*. [A&A](#), **675**, A199.
Tech Focus: Relational database architecture, data standardization, and public web portal development.
- **Avakyan A.**, Lipunova, G., & Malanchev, K. (2024). *"The effect of thermal winds on the outbursts evolution of LMXB systems"*. [MNRAS](#), **527**, 3709.
Tech Focus: High-performance C++ numerical simulations and algorithm optimization (FREDDI engine).

References

- Prof. Dr. Andrea Santangelo (santangelo@astro.uni-tuebingen.de, IAAT, Tübingen, Germany)
- Dr. Victor Doroshenko (victor.doroshenko@gmail.com, Omegalambda-tec, Garching, Germany)
- Dr. Galina Lipunova (glipunova@mpifr-bonn.mpg.de, MPIFR, Bonn, Germany)
- Dr. Konstantin Malanchev (malanchev@cmu.edu, CMU, Pittsburgh, USA)